



UNIVERSITY of
RWANDA

COLLEGE EXAMINATIONS – ACADEMIC YEAR 2024 - 2025

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF SCIENCE – DEPARTMENT OF MATHEMATICS

**YEAR I: COMPUTER ENGINEERING, COMPUTER SCIENCE,
INFORMATION SYSTEMS, INFORMATION TECHNOLOGY**

FINAL EXAMINATION – SEMESTER II

MATHEMATICS FOR ENGINEERS II (MAT 1264)

DATE: / 06 /2025

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS:

1. This paper contains **TWO** sections.
 2. Section A is compulsory. Answer any **TWO** of the **THREE** questions in Section B.
 3. No written materials are allowed in the Examination Room.
 4. Write all your answers in the answer booklet provided.
 5. Do not forget to write your Registration Number.
 6. Do not write any answers on this question paper.
 7. Calculator is permitted but you cannot share it with anyone for any reason
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Question One (Compulsory) ... / 20 marks

(a) Find all second partial derivatives of $u(x, y) = 3x^4 - 9y^4 + x^3y^3$. [2 marks]

(b) locate and identify the stationary points of $f(x, y) = \frac{x}{y} + x + y$. [2 marks]

(c) An electrostatic field is given by $\vec{E} = 3(x+y)\vec{i} + 2xy\vec{j}$.
Find the direction and the curl of $\vec{E}$ at the point $(2, 2)$. [2 marks]

(d) Find the value of the line integral along the straight line joining $(-1, -1)$ and $(1, 1)$. [2 marks]

$$\int_C (2x - y)dx + xydy$$

(e) Represent the region of integration and evaluate [2 marks]

$$\int_2^3 \int_0^2 (4 + x - y) dy dx$$

(f) Find the general solution of the equation [3 marks]

$$t \frac{dx}{dt} + x = 2t$$

(g) Solve the initial value problem for $y(t)$. [3 marks]

$$y'' - 3y' - 4y = 0, y(0) = 5, y'(0) = 0$$

(h) Solve the initial value problem using the Laplace transform [4 marks]

$$\frac{dx}{dt} + 3x - 4 = 0, x(0) = 5$$

Question Two (Optional) ... / 15 marks

(a) The liquid base of an ice cream has an initial temperature of $200^\circ F$ before it is placed in a freezer with a constant temperature of $0^\circ F$. After 1 hour, the temperature of the ice-cream base has decreased to $140^\circ F$. Formulate and solve the initial-value problem to determine the temperature of the ice cream after 2 hours. [7 marks]

(b) Using the variation of parameters method, solve the initial value problem [8 marks]

$$\frac{d^2y}{dx^2} - y = 3x^2e^x, y(0) = 1, y'(0) = 2$$

Question Three (Optional) ... / 15 marks

(a) Solve the system of differential equations:

[10 marks]

$$\begin{cases} \frac{dx}{dt} = 4x + 2y \\ \frac{dy}{dt} = 3x - y \end{cases} \quad (1)$$

(b) Evaluate the following line integral using Green's theorem

[5 marks]

$$\oint_C (x^2 \vec{i} + 4xy \vec{j}) \cdot d\vec{s}$$

where C is a closed path in the form of a triangle with vertices A(0,0), B(3,0) and (3,5).

Question Four (Optional) ... / 15 marks

(a) Solve the following differential equation using the Laplace transform

[10 marks]

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} - 3x = 2t, \quad x(0) = 1, \quad x'(0) = 2$$

(b) Let $w = f(u, v)$, $u = g(x, y)$ and $v = h(x, y)$. Find $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$.

[5 marks]

.....**END**.....